



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

Artificial Intelligence [CSE-4053]

Marks: 50

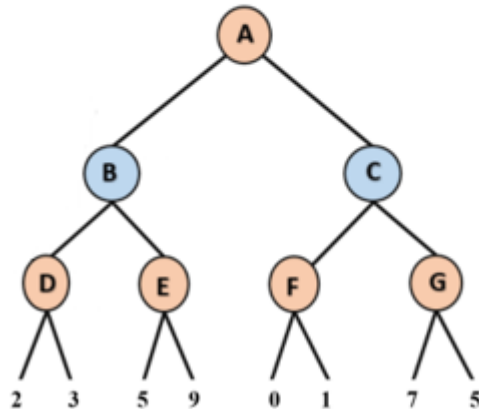
Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Differentiate between model-based agents and utility-based agents. Highlight their architecture, decision-making approach, and real-world applicability with suitable examples. (5)
 - A)
 - B) Compare and contrast between mundane task and formal task in the domain of artificial intelligence. Apply these themes and present suitable examples. (5)
- 2) Develop an agent-based solution for an Autonomous Farming System. Explain how simple reflex and model-based reflex agents can be used to handle tasks like irrigation and pest control. (5)
 - A)
 - B) Analyze the advantages and limitations of using DFS and BFS for solving AI-based search problems. Discuss their performance in shallow vs. deep search trees with suitable examples. (5)
- 3) Given the following weighted graph, use Uniform Cost Search to find the lowest-cost path from S to G. (5)
 - A) S – A (3)
S – B (1)
A – C (7)
B – C (5)
C – G (2)
B – D (9)
D – G (3)
C – E (4)
E – G (6)
 - B) (5)

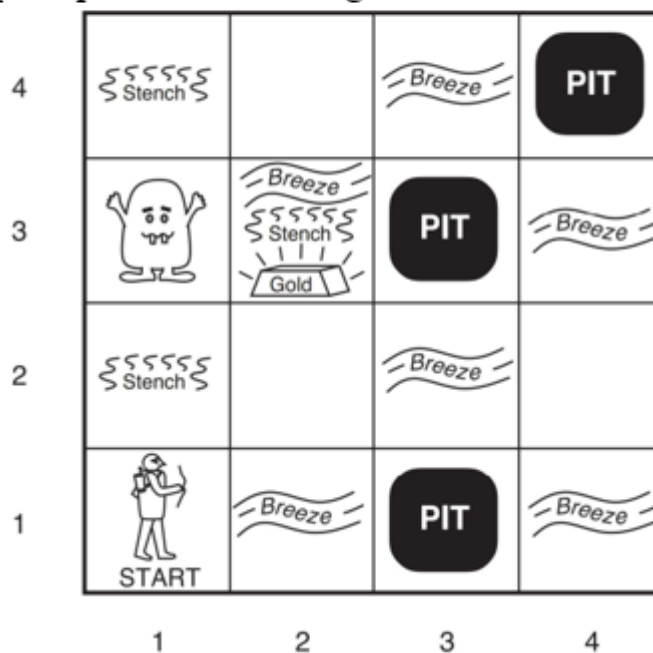
Consider the following game tree, where each leaf node is labeled and contains a utility value. Apply alpha-beta pruning to this tree. Clearly show the alpha and beta updates at each node. Mark pruned branches, if any and identify the optimal value returned by node A.



4)

Given a Wumpus World grid where the agent starts at position [1,1], evaluate and determine the safest possible path the agent can follow to reach a goal cell while avoiding pits and the Wumpus. Use percept-based reasoning.

A)



(5)

B)

Generate equivalent FOPL statements for the followings:

- i) All musicians play instruments
- ii) Some games are played online
- iii) No fruits grow in winter
- iv) Every citizen pays taxes

(5)

5)

Analyze the difference between predicate calculus and propositional logic representation.

(5)

A)

B)

A spam email classifier uses Bayesian inference with the following data:

(5)

$$P(\text{Spam}) = 0.4$$

$$P(\text{"Win"} \mid \text{Spam}) = 0.9$$

$$P(\text{"Win"}) = 0.3$$

Calculate $P(\text{Spam} \mid \text{"Win"})$ using Bayes' Theorem.

Explain the relevance of Bayesian reasoning in designing such classifiers.

-----End-----